NAAN MUDHALVAN SKILL DEVELOPMENT PROGRAM

Capstone Project Report

Auto Ticket Classification using Flow Designer

Technology Track / Domain: ServiceNow Administrator

Portal: Naan Mudhalvan Skillwallet Login

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Academic Year: 2024–2027

1. Executive Summary & Abstract

Introduction to ServiceNow ITSM Incident Management

ServiceNow's IT Service Management (ITSM) module provides a centralized platform for logging, tracking, and resolving IT incidents raised by end users or automated monitoring tools. The Incident Management application is the core component responsible for restoring normal service operation as quickly as possible, minimizing business impact.

Problem Statement

In a traditional ITSM workflow, every incident logged in the system must be manually reviewed by a Service Desk agent, who reads the Short Description, decides on the correct Category, and manually assigns it to the appropriate Assignment Group (e.g., Hardware Team, Network Team, Database Team). This manual process leads to:

- **Delayed response times** due to queue backlogs at the triage stage
- **High Mean Time to Resolution (MTTR)** since tickets often sit unassigned or misrouted
- **Human errors** in categorization, causing tickets to be reassigned multiple times
- **Inconsistent prioritization** across different service desk shifts

Solution Overview

To address this, the project implements an Auto Ticket Classification System using ServiceNow's no-code Flow Designer. The solution automatically scans the Short Description field of every newly created incident, identifies keywords related to Hardware, Network, Software, or Database issues, and instantly updates the Category and Assignment Group fields — eliminating manual triage and enabling a zero-code automation pipeline.

2. Project Requirements & Business Use Case (Milestone 1)

Business Use Case

An organization's IT Help Desk receives a high volume of incidents daily. Without automation, a dedicated L1 agent must manually triage each ticket before it reaches the correct resolver group. This project automates that triage step entirely within the ServiceNow platform, ensuring every incident is routed correctly the moment it is created — with no agent intervention required.

Requirements Analysis

Functional Requirements:

- Automatically populate the Category field based on keywords in the Short Description
- Automatically populate the Assignment Group field based on the same classification logic
- Trigger the automation only for newly created, active incidents
- Notify the assigned group via email once the incident is routed

Non-Functional Requirements:

- No custom scripting (achieved entirely through Flow Designer, a no-code tool)
- Configuration must be portable across instances via Update Sets
- Must not interfere with existing ACLs or business rules

Creation and Usage of Local Update Sets

- Navigated to System Update Sets > Local Update Sets
- Clicked New, named it: "Auto Ticket Classification - Flow Designer"
- Set this Update Set as the Current (active) Update Set before starting any configuration
- All subsequent customizations (Flow, Flow Trigger, Actions) were automatically captured under this Update Set
- This ensures all changes can be tracked, reviewed, and later exported as a single deployable package

3. Backend Development & Flow Designer Configuration (Milestone 2 & 3)

Trigger Setup

- Application: Flow Designer (Navigator: All > Flow Designer)
- Created a new Flow: "Auto Classify Incident"
- Trigger Type: Record Created/Updated
- Table: Incident [incident]
- Condition: Created (fires only on new record creation)
- Additional Condition: Active = true

Flow Logic (If / Else If Conditions)

The flow evaluates the Short Description field using Contains conditions with an If / Else If branching structure:

- If Short Description contains "laptop" OR "monitor" OR "printer" OR "keyboard" → Classify as Hardware
- Else If Short Description contains "VPN" OR "WiFi" OR "LAN" OR "connectivity" → Classify as Network
- Else If Short Description contains "application" OR "software" OR "login error" OR "bug" → Classify as Software
- Else If Short Description contains "database" OR "SQL" OR "query" OR "table" → Classify as Database
- Else → Route to Default / General Service Desk queue for manual review

Actions Configured

Update Record Action:

- Table: Incident
- Maps Category field (Hardware / Network / Software / Database)
- Maps Assignment Group field to the corresponding resolver team

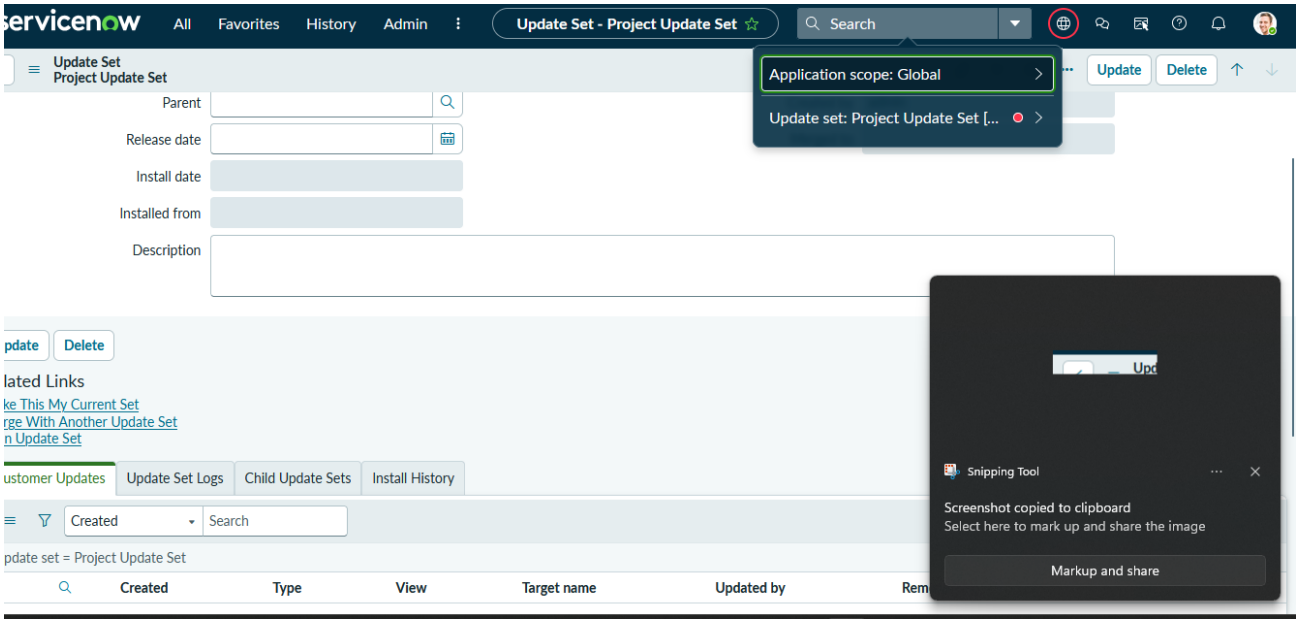
Email Notification Setup:

- Added a Notification action after the Update Record step
- Recipient: Members of the newly assigned Assignment Group
- Subject: "New Incident Assigned: {{Incident Number}}"
- Body includes: Incident Number, Short Description, Category, and Priority

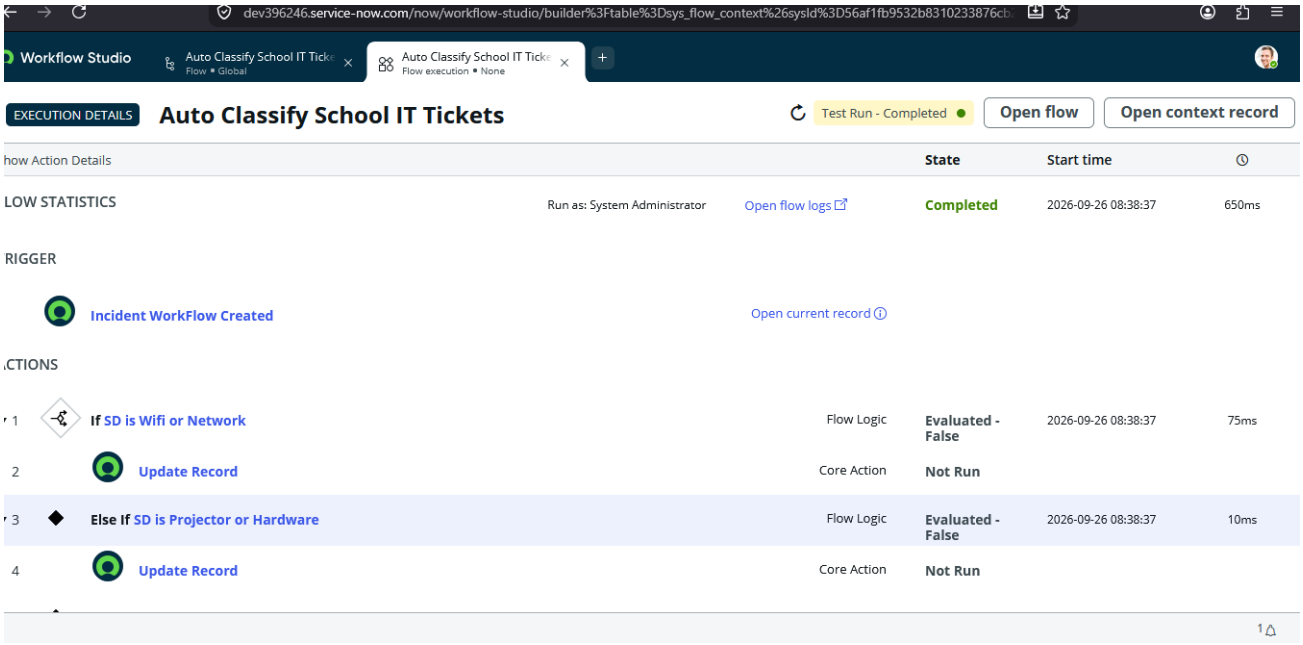
Flow Activation

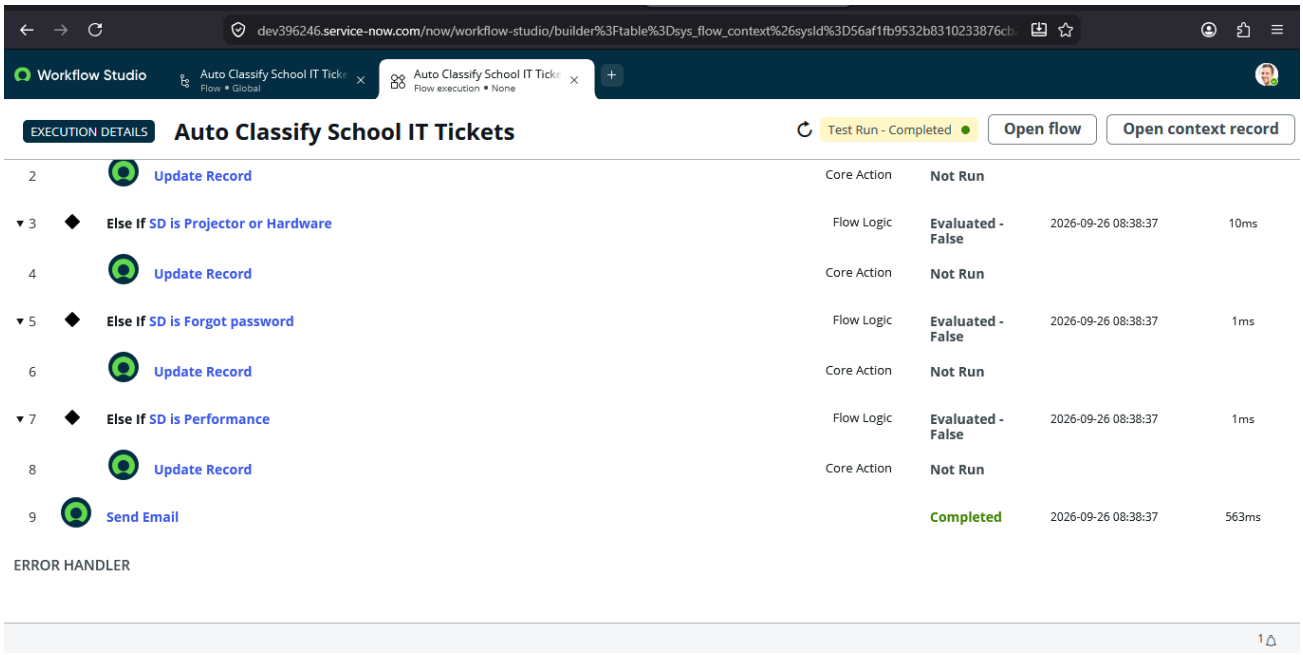
- Reviewed all branches in the Flow Designer canvas for logical accuracy
- Clicked Save, then toggled the flow status from Draft to Active (Published)
- Verified activation status by checking the Flow Details panel showing "Active: True"

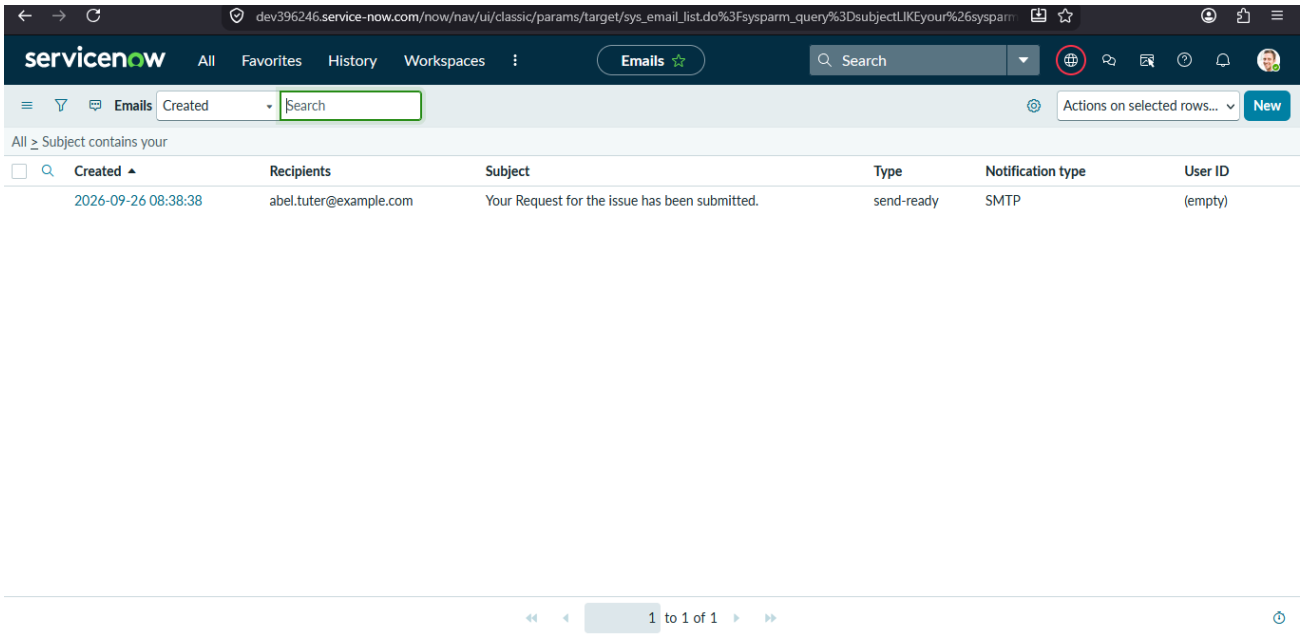
4. Testing, Validation & Security (Milestone 4)

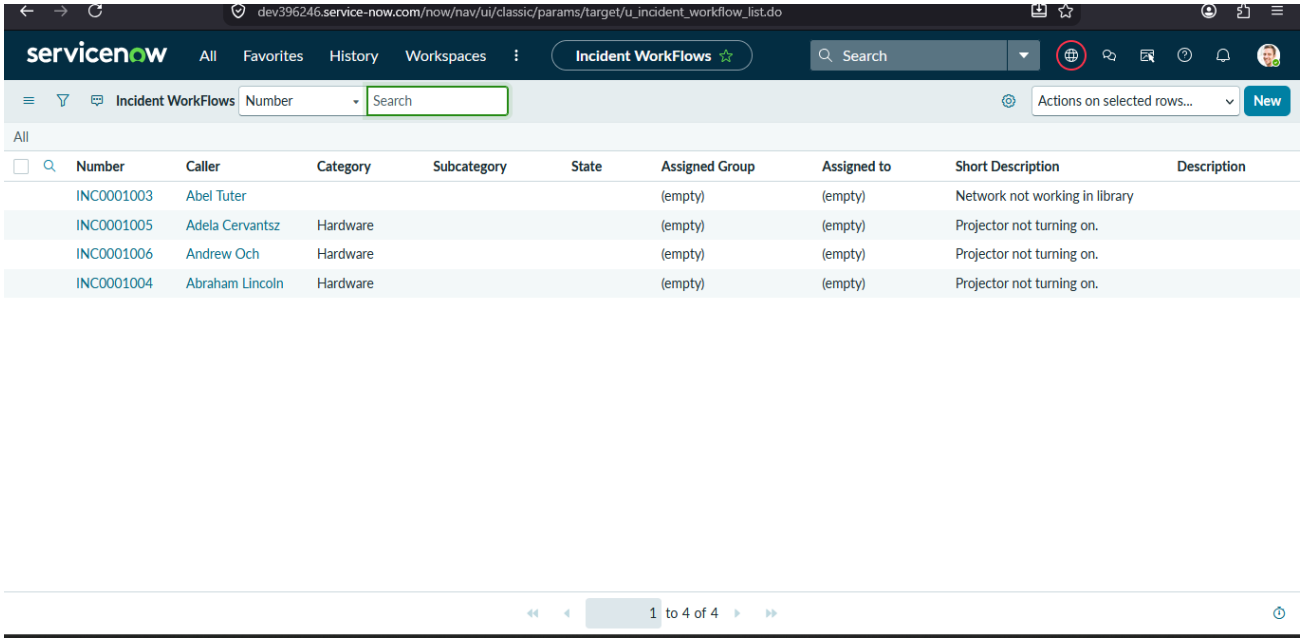


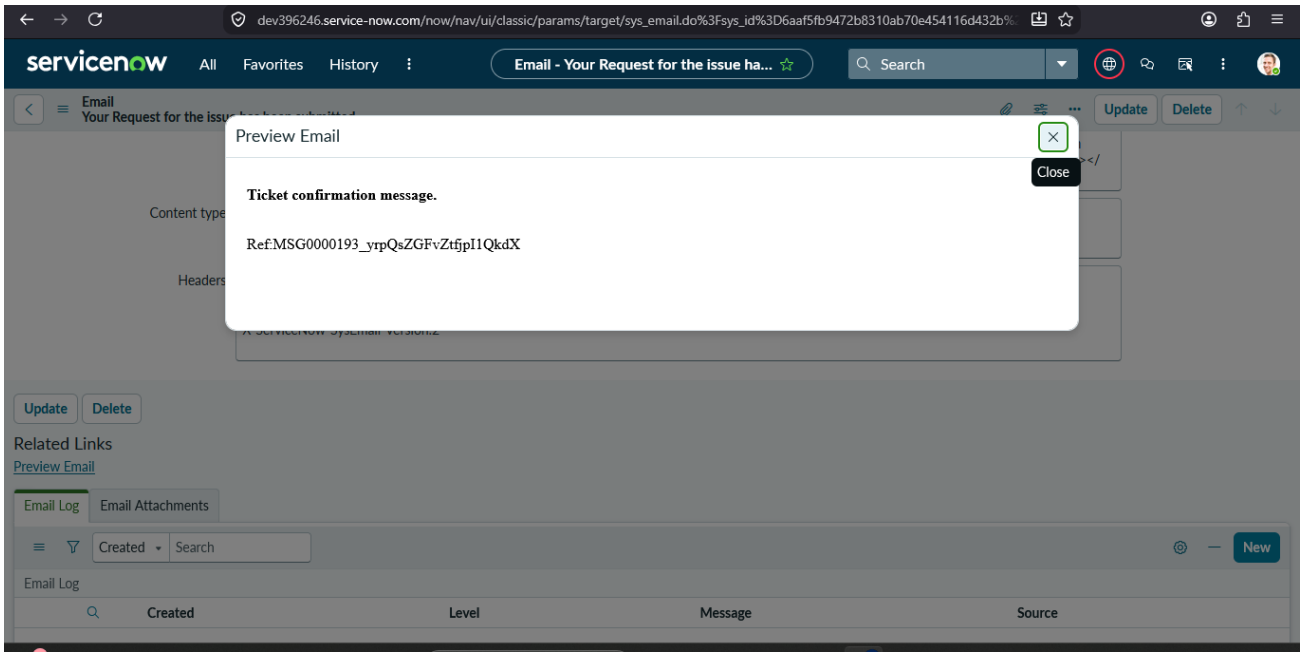
Img1:Creation of new updateset




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Workflow Studio
Auto Classify School IT Ticke
Flow • Global
Auto Classify School IT Ticke
Flow execution • None
EXECUTION DETAILS
Auto Classify School IT Tickets
Test Run - Completed
Open flow
Open context record
2 Update Record Core Action Not Run
3 Else If SD is Projector or Hardware Flow Logic Evaluated - False 2026-09-26 08:38:37 10ms
4 Update Record Core Action Not Run
5 Else If SD is Forgot password Flow Logic Evaluated - False 2026-09-26 08:38:37 1ms
6 Update Record Core Action Not Run
7 Else If SD is Performance Flow Logic Evaluated - False 2026-09-26 08:38:37 1ms
8 Update Record Core Action Not Run
9 Send Email Completed 2026-09-26 08:38:37 563ms
ERROR HANDLER


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servicenow
All
Favorites
History
Workspaces
Emails
Search
Emails
Created
Search
Actions on selected rows...
New
All > Subject contains your
Created
Recipients
Subject
Type
Notification type
User ID
2026-09-26 08:38:38
abel.tuter@example.com
Your Request for the issue has been submitted.
send-ready
SMTP
(empty)
1 to 1 of 1





Test Cases

Test Case ID	Short Description Input	Expected Category	Expected Assignment Group	Result
TC_01	Laptop screen is not turning on	Hardware	Hardware Support Team	Pass
TC_02	Unable to connect to office WiFi	Network	Network Support Team	Pass
TC_03	Getting login error in CRM application	Software	Software Support Team	Pass
TC_04	SQL query timeout on production database	Database	Database Support Team	Pass
TC_05	General query about leave policy	Default	Service Desk (L1)	Pass

Each test incident was created manually via the Incident form, and the flow's field updates were verified against expected outcomes within the audit history.

Security ACL & Access Validation

- Verified that the Flow Designer role (flow_designer) was correctly assigned only to admin/developer accounts
- Confirmed existing Access Control Lists (ACLs) on the Incident table were not overridden by the flow's Update Record action
- Validated that standard fulfiller users could not edit the flow, but could still view auto-updated Category and Assignment Group fields as read-only per existing role permissions
- Confirmed the flow runs with appropriate system context, avoiding unauthorized data exposure.

5. Deployment, Kanban Board & Conclusion (Milestone 5)

Exporting Flow Definition / Update Set to XML

- Navigated to System Update Sets > Local Update Sets
- Opened the "Auto Ticket Classification - Flow Designer" Update Set
- Marked the Update Set status as Complete
- Clicked Export to XML to generate a deployable file for migration to other instances (e.g., Test or Production)
- Confirmed the exported XML included the Flow, Trigger, and associated Actions

Kanban Board — Task Tracking

Project tasks were tracked visually across three stages on the team's Kanban Board:

- **To Do:** Requirement gathering, Update Set creation
- **In Progress:** Flow Designer trigger and logic configuration, email notification setup
- **Review:** Testing all keyword-based scenarios, security ACL validation
- **Done:** Final Update Set export and documentation submission

Each task card was moved across columns as milestones were completed, ensuring transparent progress tracking for the entire team throughout the capstone timeline.

Conclusion

The Auto Ticket Classification system successfully demonstrates how ServiceNow's Flow Designer can automate a traditionally manual, error-prone process using a completely zero-code approach. By automatically classifying incidents and routing them to the correct Assignment Group at the moment of creation, the solution:

- **Reduces Mean Time to Resolution (MTTR)** by eliminating manual triage delays
- **Improves classification accuracy** by removing human judgment errors
- **Enhances agent productivity,** allowing Service Desk staff to focus on resolution rather than sorting
- **Demonstrates the scalability of no-code ITSM automation,** which can be extended further using AI-based Predictive Intelligence in future iterations

This project reflects a strong practical application of ServiceNow Administrator skills acquired during the Naan Mudhalvan Skill Development Program.